

NEERAJ MUDUNURU

Security Researcher

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PROFESSIONAL SUMMARY

Cyber security researcher and hands-on experience in digital forensics, incident response, SOC operations, and AI-driven security tooling. Experienced in Security Monitoring, CDR/IPDR analysis, mobile forensics, malware fundamentals, and vulnerability scanning. Proven ability to coordinate security teams, communicate findings to business stakeholders, and deliver security tools. Targeting roles in Malware Analyst and Digital Forensics with government agencies or intelligence organizations.

CORE EXPERTISE

- Digital Forensics & Cybercrime Investigation
- CDR/IPDR Analysis & Evidence Handling
- SOC Operations & Log Analysis
- AI-Driven Security Systems (ML, RAG, NLP)
- Vulnerability Assessment & DAST

INTERNSHIP EXPERIENCE

Cyber Law Technical Consultant — SIRI Law LLP June 2026 – Present

- Provided technical analysis of cyber security incidents and digital evidence, supporting legal cases involving data breaches and cybercrime.
- Assisted in mapping technical findings to legal frameworks (e.g., IT Act, DPDP Act 2023), bridging the gap between cyber security and law
- Conducted security assessments and risk analysis, helping identify vulnerabilities and recommend mitigation strategies for clients

Cyber Security Engineer Trainee — COE Security LLC (North Carolina, USA – Night shifts) December 2025 – May 2026

- Handled end-to-end client onboarding for 5+ security engagements, coordinating requirements between clients and internal teams, improving project initiation efficiency and reducing onboarding delays.
- Conducted extensive OSINT research targeting key stakeholders and certification bodies, delivering intelligence reports that directly supported business development and pre-engagement planning activities.
- Maintained comprehensive project documentation — spanning client requirements, workflow updates, and progress logs — ensuring operational transparency and improving cross-team coordination efficiency.
- Enhanced an AI-based penetration testing tool by researching remote IoT attack surfaces, implementing security hardening recommendations, and configuring secure VPN tunnels for controlled client-side testing.

Agentic AI Intern — Innomatics Research Labs February 2026 – April 2026

- Engineered a GenAI-powered resume screening system using LangChain pipelines, automating candidate shortlisting and reducing manual screening effort by ~70%; integrated LangSmith for end-to-end LLM observability, tracing, and performance monitoring to improve evaluation consistency and debugging.
- Developed a production-ready RAG-based CRM prototype using Retrieval-Augmented Generation, reducing AI hallucinations and improving contextual accuracy for client-facing query resolution.
- Applied NLP fundamentals including sentiment analysis and text normalization to build an end-to-end AI chatbot pipeline with structured input processing and dynamic response generation logic.
- Delivered a FastAPI e-commerce backend with REST API integration and solved 100+ HackerRank challenges, reinforcing strong Python engineering and scalable API design skills.

Cyber Security Mentor — Cyber Security & Digital Forensics UK Pvt Ltd October 2025 – January 2026

- Directed and mentored a 5 member cyber security team across the full project lifecycle — from planning and task allocation to milestone delivery — ensuring on-time completion and high-quality outputs.
- Delivered 10+ executive-level presentations outlining project progress, threat findings, and security outcomes to senior stakeholders, enabling informed decision-making.
- Provided hands-on technical coaching to team members on cyber security tools, methodologies, and problem-solving frameworks, accelerating skill development and team performance across all engagements.

Cyber Crime Investigative Intern — Cyber Crime Department, AP Police July 2025 – August 2025

- Performed in-depth CDR and IPDR analysis on live cybercrime cases, mapping communication timelines and identifying suspicious behavioral patterns that guided active police investigations.
- Leveraged Hancm Forensic Tools and Oxygen Mobile Forensics to extract, preserve, and analyze digital evidence from mobile devices, maintaining strict chain-of-custody standards for legal admissibility.
- Investigated a fake online betting application fraud case — traced digital footprints, analyzed operational patterns, and delivered forensic findings that directly contributed to successful case resolution.
- Collaborated with law enforcement analysts to produce structured forensic reports, translating technical evidence into legally sound documentation for use in judicial proceedings.

- Executed hands-on reconnaissance and vulnerability scanning on real-world cybercrime case studies, identifying attack vectors and producing detailed threat analysis reports for investigative teams.
- Operated FTK Imager and FTK Toolkit to perform forensic evidence acquisition, media imaging, and artifact analysis while upholding chain-of-custody documentation across all case assignments.
- Achieved 92% score and Top 50 ranking in a Capture The Flag (CTF) competition, demonstrating strong applied skills in offensive security, reverse engineering, and cryptographic analysis.

PROJECTS

Aysal-Shield — Apache Tomcat DAST Scanner with CI/CD Integration (github.com/Rolex-1905/aysal-shield)

- Engineered an automated DAST scanner targeting 15+ Apache Tomcat CVEs, integrating security gates into GitHub Actions and Azure DevOps pipelines to enforce vulnerability checks on every deployment.
- Designed a CLI-first, CI/CD-native workflow enabling seamless integration of dynamic security scans into pipelines for continuous vulnerability detection.
- Eliminated manual pre-production security testing by automating scan execution and report generation, enabling development teams to detect and remediate critical web server vulnerabilities before release.

Aysal-Scan — Secret Detection & Blast Radius Analysis Tool (github.com/Rolex-1905/aysal-scan)

- Built a secret scanning tool detecting **17+ credential types** across codebases, Git history, and repositories using regex + entropy analysis
- Developed a **blast radius engine** to validate leaked keys via API calls and enumerate real permissions (AWS IAM, GitHub scopes)
- Implemented **risk scoring with actionable remediation** and SARIF integration for GitHub Security and CI/CD pipelines

Aysal-IDS — AI-Powered Network Intrusion Detection System (github.com/Rolex-1905/Aysal-IDS)

- Trained a supervised ML pipeline using Random Forest and Decision Tree classifiers on the CICIDS2017 dataset to detect and classify network intrusions — achieving 95%+ accuracy across DoS, Probe, R2L, and U2R attack categories.
- Developed an AI-based network intrusion detection system to identify malicious traffic patterns including DDoS, reconnaissance, and brute-force attacks
- Delivered 99.9%+ detection accuracy on 2.8M+ network flows, enabling high-confidence threat detection aligned with SOC monitoring workflows

GenAI Resume Screening System (github.com/Rolex-1905/ai-resume-screening-system)

- Designed an end-to-end GenAI pipeline that semantically parses and ranks resumes against job requirements using large language models, automating candidate evaluation with structured scoring and justification outputs.
- Reduced hiring friction by generating ranked shortlists with criteria-level justification, improving selection accuracy and eliminating inconsistencies in manual resume review processes.

RAG-Based CRM Query Support System with HITL (github.com/Rolex-1905/rag-customer-support-assistant)

- Architected a Retrieval-Augmented Generation CRM system that grounds all AI responses in a domain-specific knowledge base, drastically reducing hallucinations and improving factual accuracy for client-facing support queries.
- Embedded a Human-in-the-Loop (HITL) feedback loop to continuously refine retrieval quality and model outputs, enabling iterative accuracy improvements with minimal engineering overhead over time.

TECHNICAL SKILLS

Security Operations: SOC Monitoring, Threat Detection, Incident Response, Log Analysis, Network Traffic Monitoring, DAST Scanning

Digital Forensics: Mobile Forensics, CDR/IPDR Analysis, Digital Evidence Acquisition, FTK Imager, FTK Toolkit, Hancom Forensic Tools, Oxygen Mobile Forensics, LSB Steganography

Malware & Vulnerability: Malware Analysis Fundamentals, Vulnerability Scanning, Reconnaissance, SCADA/ICS Security, AI-Based Attack & Defence

Development & AI: Python, FastAPI, Generative AI, RAG, NLP, Sentiment Analysis, CI/CD (GitHub Actions, Azure DevOps)

ACHIEVEMENTS

- Ranked Top 10 in TCM Security AI Hacking CTF — competitive challenge focused on AI-based attack and defence techniques.
- Investigated and resolved a real-world cybercrime case involving a fake online betting application through digital forensics analysis at AP Police Cyber Crime Department.
- Contributor to Open Source Connect Global 2026 (OSCG 2026) — Active GitHub contributor with security tools and automation projects

EDUCATION

Bachelors of Technology in Cyber Security

2023 – 2027

Vignan's Institute of Information Technology, Visakhapatnam CGPA: 8.80/10

LANGUAGES

English (Intermediate – B2) **Telugu** (Upper Intermediate – B2)